

Pranshav Lakhia

+1 (908) 503-1754 | pranshavlakhia13@gmail.com | github.com/lakhia13 | linkedin.com/in/pranshav-lakhia

EDUCATION

The Pennsylvania State University

Bachelor of Science in Computer Science, Minor in Mathematics

Graduation — May 2026

- GPA: 3.87/4.0 | Dean's List Fall 2024, Spring 2025, Undergraduate Research Conference Grant (Fall 2025, Spring 2026))

EXPERIENCE

Software Developer

Aug 2025 — Present

Penn State University, Kim Lab

- Collaborated with a team of 4 to build CortexMap, a full-stack neuroscience research platform enabling semantic search over brain atlas literature via a custom boolean query engine, PubMed data pipeline, and LLM-assisted retrieval system
- Engineered a RAG pipeline using Python and pgvector, implementing document chunking and tokenization for PubMed abstracts to generate and index embeddings, enabling high-accuracy semantic search over neuroscience literature
- Designed gRPC service definitions to expose the RAG query engine to the TypeScript frontend, establishing a strongly-typed, low-latency communication layer

Undergraduate Research Assistant & Mentor

June 2025 — Present

D3M Labs, EnvironMentor Program 2025, Penn State University

- Mentored two high school students in quantitative research into Carbon emissions during Covid19 lockdowns; led weekly advising sessions ensuring methodological rigor
- Collaborated on multiphase sleep study with 300 participants; developed reproducible ML pipeline using SVM, XGBoost, and CNN (92% accuracy); discovered correlation between sleep characteristics and cognitive efficiency; presented research at NRCP 2025 (UPenn)
- Designed and executed controlled emotion-elicitation experiment using curated stimuli and EmotiBit sensors; processed biosignals (EDA, PPG, temperature, motion) for feature extraction

Research Assistant

Aug 2025 — December 2025

Penn State University, Dr. Hyuntae Na's Group

- Conducted intensive literature review of computational biology publications relating to protein-protein interfaces and binding free energy
- Developed preprocessing pipeline to generate 3d structures and visualizations using PyMol for feature analysis from raw PDB files, including contact maps, electrostatic potential maps, and residue interaction networks
- Trained graph neural networks using PyTorch Geometric and DeepRank2 to predict binding free energy with 94% accuracy, presented said research at NCRC 2026 at Harvard University

PROJECTS

Solo Developer, Opportunist - AI driven Job Scraper

github.com/lakhia13/opportunist

- Developed Python-based AI-driven opportunity aggregator crawling career, scholarship, and research portals with daily digest delivery
- Engineered system with Scrapy/Playwright web crawlers, MongoDB Atlas backend, OpenAI/SentenceTransformer embeddings, and SendGrid email delivery
- Deployed via Docker/Heroku with free-tier optimization strategies and modular micro-services architecture

Solo Developer, Lepa - Toy Theorem Prover Language

github.com/lakhia13/lepa

- Designed and implemented LEPA theorem-prover language in ASCII syntax supporting propositional logic, predicate logic, and set theory
- Built transpiler in Java using JFlex and CUP with complete parsing and code-generation pipeline
- Documented extension pathways to scale LEPA into teaching tool for high-school logic instruction

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Java, Rust, SQL (Postgres, SQLite), JavaScript, Typescript, Matlab
- **Frameworks & Tools:** Numpy, Pandas, Matplotlib, Tensorflow, Pytorch, Docker, React, Flask, Vercel, Docker